

# Research Report on Cast Aluminum Alloys for BEV Auxiliary Motor Components

---

**Date:** 2026-03-17

## Executive Summary

---

The rapid expansion of the Battery Electric Vehicle (BEV) market has intensified the focus on optimizing component design for efficiency, performance, and cost. Auxiliary motors, which power systems such as pumps, compressors, and steering, are critical to overall vehicle function. This report provides a comprehensive analysis of cast aluminum alloys used in the components of these motors, such as housings, end covers, and structural brackets. The selection of these materials is driven by a strategic balance of properties essential for automotive applications, including low density for vehicle light-weighting, high thermal conductivity for effective heat management, and robust mechanical strength.

This report details the elemental compositions, mechanical and physical properties, and primary manufacturing processes for a range of prevalent aluminum alloys. Key alloys analyzed include the widely used Al-Si-Cu family, such as A380, A383, and EN AC-46000 (LM24), which are favored for high-pressure die casting due to their excellent fluidity and suitability for mass production of complex parts [7, 19, 31]. The report also examines the Al-Si-Mg family, including A356, A357, and AlSi10Mg, which are often heat-treated to achieve superior strength and ductility [13, 14, 25]. These alloys are frequently processed using permanent mold casting to produce components with high structural integrity and pressure tightness, essential for liquid-cooled motor applications [16, 28].

A comparative analysis of high-pressure die casting and permanent mold casting reveals a fundamental trade-off between production volume, part complexity, and material integrity. High-pressure die casting is optimal for high-volume, intricate components where its speed and precision outweigh the higher tooling costs and potential for internal porosity [16, 28, 42]. In contrast, permanent mold casting offers a path to mechanically superior, less porous parts at lower tooling costs, making it suitable for moderate production volumes and components where strength and pressure tightness are paramount [16, 28, 29]. The final selection of an alloy and casting method is a critical engineering decision that directly impacts the performance, reliability, and cost of BEV auxiliary motors.

## Introduction

---

The global automotive industry is undergoing a transformative shift towards electrification, with Battery Electric Vehicles (BEVs) at the forefront of this revolution. This transition places immense pressure on engineers to reimagine and optimize every vehicle system to maximize efficiency, extend driving range, and ensure long-term reliability. Auxiliary motors, while smaller than the main traction motor, play a vital role in vehicle operation, powering essential functions like climate control compressors, coolant pumps, power steering systems, and braking actuators. The performance and efficiency of these subsystems are integral to the vehicle's overall energy consumption and passenger comfort. Consequently, the materials used to construct these motors are subject to intense scrutiny.

Cast aluminum alloys have emerged as the materials of choice for many auxiliary motor components, primarily due to their compelling combination of low density, high thermal conductivity, good mechanical properties, and manufacturability. The use of aluminum directly supports the industry-wide goal of lightweighting, where reducing vehicle mass leads to lower energy consumption and increased range.

Furthermore, the operational efficiency of electric motors and their integrated power electronics generates significant heat, making the superior thermal conductivity of aluminum critical for effective thermal management and the prevention of overheating.

This report aims to provide a comprehensive research analysis of the cast aluminum alloys predominantly used in the manufacture of BEV auxiliary motor components. The scope of this investigation includes a detailed examination of the elemental composition, mechanical and physical properties, and common applications of key alloys such as A380, A356, A319, and their international equivalents. The report will also provide a comparative analysis of the primary manufacturing processes—high-pressure die casting and permanent mold casting—elucidating how the choice of process influences material properties and component design. By synthesizing this information, the report will deliver a clear rationale for why specific alloys are selected for motor applications, balancing the competing demands of performance, cost, and high-volume production.

## **The Role of Aluminum Alloys in BEV Motor Components**

---

The selection of materials for BEV auxiliary motor components is a strategic decision governed by the fundamental requirements of electric powertrain systems: efficiency, thermal stability, and durability within a compact, lightweight package. Cast aluminum alloys have become indispensable in this context, offering a suite of properties that directly address these challenges. Their primary contribution is to vehicle lightweighting. With a density approximately one-third that of steel, aluminum alloys enable the production of motor housings, brackets, and end covers that significantly reduce the overall mass of the auxiliary motor assembly [7]. This weight reduction, when aggregated across multiple vehicle systems, contributes to lower overall vehicle weight, which in turn reduces energy consumption and helps extend the BEV's driving range—a paramount concern for consumers and manufacturers alike.

Beyond lightweighting, the thermal management of electric motors is a critical factor for performance and longevity. The flow of electric current through motor windings and the operation of integrated power electronics generate substantial amounts of heat. If not dissipated effectively, this heat can degrade winding insulation, reduce the efficiency of electronic components, and ultimately lead to premature motor failure. Aluminum alloys possess excellent thermal conductivity, typically ranging from 96 to 160 W/m·K depending on the specific alloy [7, 13, 24]. This property allows motor housings and integrated heat sinks to act as effective conduits, drawing heat away from critical internal components and dissipating it to the surrounding environment or a liquid cooling system. Alloys like A380 and AlSi10Mg are particularly valued for this characteristic, enabling compact motor designs that can operate at high power densities without thermal compromise [7, 25].

Furthermore, auxiliary motor components must possess sufficient mechanical strength and stiffness to maintain dimensional stability under operational loads. The motor housing must securely support bearings, maintain precise alignment between the rotor and stator, and withstand vibrations and shocks from the vehicle's operation. Heat-treatable alloys such as A356 and A357, when subjected to a T6 temper, can achieve tensile strengths exceeding 276 MPa and 350 MPa, respectively, providing the structural integrity required for demanding applications [13, 14, 24, 26]. This high strength-to-weight ratio ensures that components can be both strong and light. Finally, the economics of mass production heavily influence material choice. Aluminum alloys are exceptionally well-suited for high-volume casting processes, which allow for the creation of complex, near-net-shape parts with integrated features like cooling fins, mounting bosses, and fluid channels [16]. This reduces the need for subsequent machining operations, lowering manufacturing complexity and overall component cost, a critical factor in the competitive automotive market.

# Analysis of Common Casting Methods for Motor Components

---

The manufacturing process used to form an aluminum motor component is as critical as the alloy selection itself, profoundly influencing the part's final properties, cost, and suitability for a given application. For BEV auxiliary motor components, the two most prevalent methods are high-pressure die casting and permanent mold casting. The choice between them involves a careful evaluation of production volume, geometric complexity, required mechanical performance, and budget constraints.

## High-Pressure Die Casting (HPDC)

High-pressure die casting is a highly automated process where molten aluminum is injected into a hardened steel mold, known as a die, under immense pressure. This pressure, which can exceed 30,000 psi, forces the metal into every detail of the mold cavity at high velocity, enabling the production of parts with exceptional dimensional accuracy and very thin walls [18]. The process is characterized by extremely fast cycle times, making it the preferred method for high-volume production runs typical of the automotive industry [16, 28]. The rapid solidification against the cool die walls results in a fine-grained surface structure, yielding an excellent as-cast surface finish that often requires no further cosmetic treatment [16, 28]. This method is ideal for producing complex and intricate motor housings with integrated features such as cooling fins, thin ribs, and complex internal geometries, minimizing the need for secondary assembly or machining operations.

However, the high-pressure die casting process is not without its disadvantages. The high velocity and turbulence of the metal injection can trap air and gases within the die, leading to the formation of internal porosity [16, 42]. While this porosity may not affect the external appearance or basic function of a part, it can compromise its structural integrity and pressure tightness [28]. For motor housings that are part of a liquid cooling circuit, this porosity can create leak paths, making the component unsuitable without costly impregnation treatments. Furthermore, the steel dies are complex and expensive to manufacture, representing a significant upfront investment [16, 29]. This high tooling cost makes HPDC economically unviable for low- to moderate-volume production. Alloys commonly used in HPDC, such as A380 and LM24, are selected for their superb fluidity, which allows them to fill these intricate molds effectively [7, 19, 22].

## Permanent Mold Casting

Permanent mold casting, which includes gravity and low-pressure variants, offers a distinct alternative to HPDC. In this process, molten aluminum is poured into a reusable steel or cast iron mold, filling the cavity under the force of gravity or with the assistance of low gas pressure [16, 42]. The metal flow is significantly less turbulent than in HPDC, which allows air and gases to escape through vents in the mold. This results in a casting with substantially lower internal porosity, leading to a denser, more uniform microstructure [28, 42]. Consequently, permanent mold castings typically exhibit superior mechanical properties, including higher tensile strength, greater ductility, and improved pressure tightness compared to their die-cast counterparts [16, 28]. This makes the process highly suitable for structurally critical components or for motor housings that must contain liquid coolants without leaking.

The primary trade-offs for these improved material properties are a slower production rate and limitations on geometric complexity. The cycle times for permanent mold casting are considerably longer than for HPDC, which can increase the per-unit cost for very large production runs [16, 28]. Additionally, because the process relies on gravity or low pressure to fill the mold, it is less suited for producing parts with extremely thin walls or the same level of intricate detail achievable with HPDC [42]. However, the tooling for permanent mold casting is generally simpler and less expensive than for die casting, making it a more economical choice for moderate production volumes [16, 29]. Heat-treatable

alloys like A356 and A357 are frequently used in permanent mold casting, as the low-porosity structure responds exceptionally well to heat treatment, enabling the production of high-strength, high-integrity components for demanding motor applications [8, 14].

## Detailed Analysis of Key Cast Aluminum Alloys

The diverse operational requirements of BEV auxiliary motor components have led to the adoption of a variety of cast aluminum alloys, each with a unique profile of composition and properties. These alloys can be broadly categorized based on their primary alloying elements, which dictate their performance in casting, their mechanical characteristics, and their response to environmental and thermal stresses. The most prominent families are the Aluminum-Silicon-Copper (Al-Si-Cu) alloys, known for their excellent castability in die casting, and the Aluminum-Silicon-Magnesium (Al-Si-Mg) alloys, prized for their high strength and response to heat treatment.

### Al-Si-Cu Alloys: The Workhorses of Die Casting

This family of alloys is defined by significant additions of both silicon and copper and is the cornerstone of high-pressure die casting for automotive components. Silicon is added to enhance fluidity and reduce solidification shrinkage, while copper contributes significantly to hardness and strength, particularly at elevated temperatures.

**A380, EN AC-46000 (AlSi9Cu3(Fe)), and LM24** represent the most widely used alloys in this category. They are often considered interchangeable in many applications [31]. Their composition typically includes 8-11% silicon and 2-4% copper [1, 7, 19, 35]. This formulation provides an exceptional balance of properties, making them a default choice for many motor components. They exhibit excellent castability, allowing them to fill the complex geometries of motor housings and end plates with high fidelity. Their thermal conductivity, around 96-120 W/m·K, is sufficient for many air-cooled or moderately loaded motor applications [7, 35]. Mechanically, A380 provides a typical as-cast tensile strength of about 320 MPa and a Brinell hardness of 80 HBW, offering good structural performance for housings and brackets [7, 31]. However, the copper content renders these alloys susceptible to corrosion, often requiring protective coatings if exposed to harsh environments. Their primary application in BEV auxiliary motors is for mass-produced, complex-shaped housings where the combination of low cost, high production speed, and good all-around properties is paramount.

**A383 (ADC12)** is a modification of A380, designed to offer even better die-filling characteristics [6]. With a higher silicon content (9.5-11.5%) and slightly lower copper (2.0-3.0%), A383 exhibits superior fluidity and an increased resistance to hot cracking during solidification [6, 18, 41]. This makes it particularly suitable for producing components with very thin walls or highly intricate features that might be challenging to cast with A380. While its mechanical properties, such as a tensile strength of 310 MPa, are very similar to A380, its primary advantage is in manufacturability [6, 18]. For BEV auxiliary motors, A383 is selected for housings that push the limits of design complexity, ensuring complete mold fill and reducing casting defects in challenging geometries.

**A319** is another important Al-Si-Cu alloy, characterized by a lower silicon content (5.5-6.5%) and a high copper content (3.0-4.0%) [9, 33]. This composition provides a good combination of casting and machining characteristics. While its fluidity is not as high as that of the A380 family, it offers good strength and is responsive to heat treatment [9, 33]. An A319 casting in the T6 condition can achieve a tensile strength of 250 MPa [9, 33]. Its good machinability makes it a strong candidate for components that require significant post-casting finishing, such as precision-machined mounting surfaces or bearing bores [9]. In the context of auxiliary motors, A319 is used for parts like engine crankcases, oil pans, and structural brackets where a balance of castability, strength, and machinability is required [9, 10].

## Al-Si-Mg Alloys: High Integrity and Heat-Treatable Performance

This alloy family is distinguished by the presence of magnesium, which allows the alloys to be significantly strengthened through precipitation hardening (heat treatment). They typically contain very low levels of impurities like iron, which enhances their ductility and toughness. These alloys are the preferred choice for components requiring high strength, fatigue resistance, and pressure tightness.

**A356 and A357** are the premier alloys in this group. A356 contains 6.5-7.5% silicon and 0.25-0.45% magnesium, while A357 has a higher magnesium content of 0.40-0.70%, enabling it to achieve even greater strength [8, 14, 26]. A critical feature of these alloys is their very low iron content (often below 0.20%), which prevents the formation of brittle intermetallic phases [8, 13]. When cast using the permanent mold process to ensure low porosity and then subjected to a T6 heat treatment (solution treatment and artificial aging), these alloys develop outstanding mechanical properties. A356-T6 can reach a tensile strength of 276 MPa with 5% elongation, while A357-T6 can exceed 350 MPa [13, 14, 24, 26]. This combination of high strength and good ductility, along with excellent corrosion resistance, makes them ideal for high-performance structural applications [13, 26]. In BEV auxiliary motors, they are used for components where structural failure is not an option, such as steering knuckles or suspension components, and for liquid-cooled motor housings where the pressure tightness afforded by the permanent mold process is essential.

**AlSi10Mg** is a versatile alloy with a composition of 9-11% silicon and 0.25-0.45% magnesium [15, 25, 40]. It bridges the gap between the high-fluidity die-casting alloys and the high-strength structural alloys. It possesses good castability and, like A356, is heat-treatable [15, 25]. After a T6 heat treatment, it can achieve a tensile strength of over 250 MPa [25, 44]. Its thermal conductivity is also favorable, increasing from around 110 W/m·K in the as-cast state to over 155 W/m·K after heat treatment [25, 44]. This makes it an excellent candidate for motor components that require both structural integrity and efficient heat dissipation, such as housings for high-power-density motors. Furthermore, AlSi10Mg is one of the most common aluminum alloys used in additive manufacturing (3D printing), allowing for rapid prototyping of motor components and the production of highly optimized, complex geometries that are impossible to cast [15, 25, 44].

## High-Silicon and Specialized Alloys

A few other alloys are used for specific niche requirements in motor design. **A360**, with 9.0-10.0% silicon and very low copper ( $\leq 0.6\%$ ), offers superior corrosion resistance and better strength at elevated temperatures compared to A380 [3, 17, 45]. This makes it a suitable choice for motor components that may be exposed to corrosive road salt or operate in a high-temperature under-hood environment. **LM6**, a eutectic alloy with 10-13% silicon and very low impurities, provides exceptional fluidity and outstanding corrosion resistance [19, 30, 46]. Its primary application is for intricate, thin-walled castings, such as complex liquid-cooling jackets or marine components, where its castability and resistance to the elements outweigh its moderate strength and poor machinability [30].

## Comprehensive Alloy Comparison Table

The following table provides a comparative overview of the key cast aluminum alloys discussed in this report, summarizing their primary manufacturing method, elemental composition, and essential mechanical and physical properties relevant to BEV auxiliary motor applications.

| Alloy Designation         | Common Casting Method(s)  | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                        |
|---------------------------|---------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>A380</b>               | High-Pressure Die Casting | Si: 7.5-9.5, Cu: 3.0-4.0      | 317 - 324              | 159 - 160            | 3.5            | 80             | 96.2                         | Excellent castability, good all-around properties. The industry standard for mass-produced motor housings and brackets. [7, 31] |
| <b>EN AC-46000 / LM24</b> | High-Pressure Die Casting | Si: 8.0-11.0, Cu: 2.0-4.0     | 240 - 320              | 140 - 165            | 1.0            | 80 - 85        | 110 - 120                    | European/British equivalent to A380. Excellent for pressure die-cast parts like engine                                          |

| Alloy Designation   | Common Casting Method(s)          | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                      |
|---------------------|-----------------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|                     |                                   |                               |                        |                      |                |                |                              | components and pump housings. [1, 19, 22, 35]                                                                                 |
| <b>A383 (ADC12)</b> | High-Pressure Die Casting         | Si: 9.5-11.5, Cu: 2.0-3.0     | 310 - 325              | 150 - 160            | 3.5            | 75             | 96                           | Superior fluidity and hot crack resistance compared to A380. Ideal for intricate, thin-walled motor housings. [6, 18, 27, 41] |
| <b>A319</b>         | Permanent Mold, Sand, Die Casting | Si: 5.5-6.5, Cu: 3.0-4.0      | 185 (F) - 250 (T6)     | 125 (F) - 165 (T6)   | 2.0            | 80 - 95        | 120 - 150                    | Good balance of castability and machinability;                                                                                |

| Alloy Designation | Common Casting Method(s)     | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                         |
|-------------------|------------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                   |                              |                               |                        |                      |                |                |                              | heat-treatable. Used for motor crank-cases, housings, and manifolds. [9, 10, 12, 33]                                             |
| A356-T6           | Permanent Mold, Sand Casting | Si: 6.5-7.5, Mg: 0.25-0.45    | 276 - 310              | 207 - 234            | 5 - 12         | 80 - 100       | 130 - 160                    | Excellent strength, ductility, and corrosion resistance after heat treatment. Used for high-integrity structural parts and pres- |

| Alloy Designation | Common Casting Method(s)     | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                                                         |
|-------------------|------------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                              |                               |                        |                      |                |                |                              | sure-tight liquid-cooled housings. [8, 13, 24, 32]                                                                                                               |
| <b>A357-T6</b>    | Permanent Mold, Sand Casting | Si: 6.5-7.5, Mg: 0.40-0.70    | 350 - 380              | 260 - 270            | 4 - 8          | 85 - 120       | 120 - 150                    | Higher strength than A356 due to more Mg. Used for high-performance aerospace and automotive structural components requiring maximum strength. [2, 4, 5, 14, 26] |

| Alloy Designation | Common Casting Method(s)   | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                                                  |
|-------------------|----------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AlSi10Mg</b>   | Die Casting, Additive Mfg. | Si: 9.0-11.0, Mg: 0.25-0.45   | 250 - 300 (T6)         | 180 - 300 (T6)       | 2 - 11         | 110 - 130      | 155 - 165 (T6)               | Versatile alloy with good strength and thermal conductivity. Used for motor housings and heat sinks; popular for 3D printing prototypes. [15, 25, 40, 44] |
| <b>A360</b>       | High-Pressure Die Casting  | Si: 9.0-10.0, Cu: ≤0.6        | 317                    | 165                  | 3.5            | 75             | 113                          | Superior corrosion resistance and high-temperature strength compared                                                                                      |

| Alloy Designation | Common Casting Method(s)    | Key Alloying Elements (wt. %) | Tensile Strength (MPa) | Yield Strength (MPa) | Elongation (%) | Hardness (HBW) | Thermal Conductivity (W/m·K) | Key Characteristics & Motor Applications                                                                                                    |
|-------------------|-----------------------------|-------------------------------|------------------------|----------------------|----------------|----------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                             |                               |                        |                      |                |                |                              | to A380. Suitable for motors in harsh environments. [3, 17, 21, 45]                                                                         |
| <b>LM6</b>        | Die Casting, Permanent Mold | Si: 10.0-13.0, Cu: ≤0.1       | 160 - 280              | -                    | 5 - 18         | -              | -                            | Exceptional fluidity and corrosion resistance. Used for intricate, thin-walled castings like cooling jackets and marine parts. [19, 30, 46] |

## Selection Rationale for BEV Motor Applications

The selection of a specific cast aluminum alloy and corresponding manufacturing process for a BEV auxiliary motor component is a complex engineering decision driven by a hierarchy of performance requirements, manufacturing constraints, and economic factors. There is no single “best” alloy; rather, the optimal choice is the one that most effectively satisfies the specific demands of the component’s

function within the broader vehicle system. The rationale can be broken down into several key areas of consideration.

First and foremost is **thermal management**. The efficiency of an electric motor and its associated power electronics is intrinsically linked to their operating temperature. Alloys with high thermal conductivity are essential for components designed to dissipate heat. For a high-power-density auxiliary motor, such as an electric air conditioning compressor that runs continuously, a housing made from an alloy like AlSi10Mg (155-165 W/m·K in T6 condition) or A356 (130-160 W/m·K) would be highly advantageous [13, 24, 25, 44]. These alloys can effectively draw heat away from the stator and electronics, maintaining optimal operating temperatures. For less thermally stressed applications, the adequate thermal conductivity of workhorse die-casting alloys like A380 (96.2 W/m·K) is often sufficient, especially when combined with design features like cooling fins [7, 31].

Second is the **strength-to-weight ratio**. Motor components must provide robust structural support for rotating assemblies and withstand vibration and shock loads without failure or deformation. The housing must maintain precise alignment of bearings and the air gap between the rotor and stator. For components that also serve a structural, load-bearing role, such as integrated mounting brackets, high mechanical strength is non-negotiable. In these cases, heat-treated Al-Si-Mg alloys are the clear choice. An A357-T6 casting provides a tensile strength comparable to some steels but at a fraction of the weight, making it ideal for applications where performance cannot be compromised [5, 14, 26]. For standard housings that are not primary structural elements, the moderate but reliable strength of die-cast A380 or LM24 is perfectly adequate and more cost-effective [7, 19].

Third, **manufacturability and cost** are often the deciding factors, particularly in the high-volume automotive sector. The need to produce millions of identical components per year heavily favors the high-pressure die casting process [16, 28]. The ability to produce a complex, near-net-shape motor housing in a cycle time of a minute or less is a powerful economic driver. This makes alloys with excellent fluidity, such as A380 and A383, the default choice for the vast majority of mass-produced auxiliary motor housings [6, 7]. The high upfront cost of the die is amortized over the large production volume, resulting in a low per-unit price. Conversely, for a lower-volume BEV model or for a component requiring superior mechanical integrity, the lower tooling cost and superior material properties of permanent mold casting with an alloy like A356 become more attractive, despite the slower production speed [16, 29].

Finally, specific operational requirements such as **pressure tightness and corrosion resistance** guide the selection. For the growing number of liquid-cooled auxiliary motors, the housing must act as a sealed vessel for the coolant. The inherent risk of porosity in high-pressure die castings makes them less suitable for this role without secondary impregnation processes [16, 28]. The dense, low-porosity structure of a permanent mold casting is far superior for ensuring a leak-proof component, making alloys like A356 the standard for such designs [8, 13]. Similarly, for a motor located in an area exposed to road salt and moisture, an alloy with low copper content like A360 or LM6 would be chosen for its enhanced corrosion resistance, ensuring the long-term durability of the component [3, 19, 30].

## Conclusion

The selection of cast aluminum alloys for BEV auxiliary motor components is a sophisticated exercise in balancing competing engineering and economic priorities. The analysis reveals a clear bifurcation in material and process strategy, tailored to the specific function of the component. For the mass production of complex, general-purpose motor housings, the combination of high-pressure die casting and Al-Si-Cu alloys, particularly A380 and its equivalents like EN AC-46000, remains the dominant and most cost-effective solution. These alloys provide an exceptional blend of castability, adequate mechanical

strength, and sufficient thermal conductivity to meet the demands of a wide range of auxiliary motor applications.

For applications requiring higher performance, whether in the form of superior mechanical strength, pressure tightness for liquid cooling, or enhanced fatigue life, the focus shifts to heat-treatable Al-Si-Mg alloys like A356 and A357, typically processed via permanent mold casting. This pathway yields components with significantly lower porosity and superior material properties, justifying the slower production cycle and different cost structure for critical applications. The choice is not merely about selecting a material from a datasheet but about co-designing the component, the alloy, and the manufacturing process to achieve the optimal outcome. As BEV technology continues to evolve, with increasing power densities and greater integration of systems, the demand for high-performance, light-weight, and thermally efficient materials will only intensify, ensuring that the continued development and intelligent application of cast aluminum alloys will remain central to the future of electric mobility.

## References

---

1. [ENAC-ALSi9Cu3\(Fe\) / ENAC-46000 - SteelNumber - Aluminium equivalent, chemical composition, properties - SteelNumber](http://www.steelnumber.com/en/steel_alloy_composition_eu.php?name_id=1244) ([http://www.steelnumber.com/en/steel\\_alloy\\_composition\\_eu.php?name\\_id=1244](http://www.steelnumber.com/en/steel_alloy_composition_eu.php?name_id=1244))
2. [A357 Aluminum Alloy - Inox Cast - Inox Cast](https://www.inoxcast.com/a357-aluminum/) (<https://www.inoxcast.com/a357-aluminum/>)
3. [A360 Aluminum Alloy: Properties and Top Applications? - Gabrian International](https://www.gabrian.com/a360-aluminum-alloy/) (<https://www.gabrian.com/a360-aluminum-alloy/>)
4. [A357 Aluminum Alloy Casting | Patriot Foundry NH - Patriot Foundry](https://www.patriotfoundry.com/services/a357-alloy-casting/) (<https://www.patriotfoundry.com/services/a357-alloy-casting/>)
5. [What is A357 Aluminum Alloy - A357 Aluminum Composition, Properties, Equivalent, A356 vs A357 Alloy - Diecasting-mould](https://www.diecasting-mould.com/news/what-is-a357-aluminum-alloy-a357-aluminum-composition-properties-equivalent-a356-vs-a357-alloy) (<https://www.diecasting-mould.com/news/what-is-a357-aluminum-alloy-a357-aluminum-composition-properties-equivalent-a356-vs-a357-alloy>)
6. [A383 Aluminum Alloy: ADC12 - Gabrian International](https://www.gabrian.com/a383-aluminum-alloy-adc12/) (<https://www.gabrian.com/a383-aluminum-alloy-adc12/>)
7. [A380 Aluminum Alloy Properties - aludiecasting.com](https://aludiecasting.com/a380-aluminum-alloy-properties/) (<https://aludiecasting.com/a380-aluminum-alloy-properties/>)
8. [A356 Aluminum Alloy - newaydiecast.com](https://www.newaydiecast.com/zh/services/aluminum-die-casting/a356) (<https://www.newaydiecast.com/zh/services/aluminum-die-casting/a356>)
9. [A319 - newaydiecast.com](https://www.newaydiecast.com/services/aluminum-die-casting/a319) (<https://www.newaydiecast.com/services/aluminum-die-casting/a319>)
10. [319 Aluminum Alloy: Common Uses, Composition, & Properties - Precision Enterprises](https://precision-enterprises.com/319-alloy/) (<https://precision-enterprises.com/319-alloy/>)
11. [a319 aluminum alloy: Topics by Science.gov - Science.gov](https://www.science.gov/topicpages/a/a319+aluminum+alloy) (<https://www.science.gov/topicpages/a/a319+aluminum+alloy>)
12. [319.0 \(SC64D, A03190\) Cast Aluminum :: MakeltFrom.com - MakeltFrom.com](https://www.makeitfrom.com/material-properties/319.0-SC64D-A03190-Cast-Aluminum) (<https://www.makeitfrom.com/material-properties/319.0-SC64D-A03190-Cast-Aluminum>)
13. [A356 Aluminum Alloy - partmfg.com](https://www.partmfg.com/a356-aluminum-alloy/) (<https://www.partmfg.com/a356-aluminum-alloy/>)
14. [A357 Aluminum Properties, Casting and Composition Guide - PARTMFG](https://www.partmfg.com/a357-aluminum-alloy/) (<https://www.partmfg.com/a357-aluminum-alloy/>)
15. [AlSi10Mg - Wikipedia - en.wikipedia.org](https://en.wikipedia.org/wiki/AlSi10Mg) (<https://en.wikipedia.org/wiki/AlSi10Mg>)
16. [Die Casting vs Permanent Mold Casting - Ace-mold](https://www.ace-mold.com/die-casting-vs-permanent-mold-casting/) (<https://www.ace-mold.com/die-casting-vs-permanent-mold-casting/>)

17. [What is A360 Aluminum - A360 Cast Aluminum Properties, Composition & 6061-T6 vs A360 - DIECASTING-MOULD](https://www.diecasting-mould.com/news/what-is-a360-aluminum-a360-cast-aluminum-properties-composition-6061-t6-vs-a360) (https://www.diecasting-mould.com/news/what-is-a360-aluminum-a360-cast-aluminum-properties-composition-6061-t6-vs-a360)
18. [A383 Aluminum Alloy - langhe-industry.com](https://langhe-industry.com/a383-aluminum-alloy/) (https://langhe-industry.com/a383-aluminum-alloy/)
19. [LM6 vs LM25 vs LM24 Aluminum Alloy: What are the Differences? - Diecasting-mould](https://www.diecasting-mould.com/news/lm6-vs-lm25-vs-lm24-aluminum-alloy-what-are-the-differences) (https://www.diecasting-mould.com/news/lm6-vs-lm25-vs-lm24-aluminum-alloy-what-are-the-differences)
20. [A360 Aluminum Alloy Composition, Properties, and Applications - TexaMetals](https://texa-metals.com/blog/a360-aluminum-alloy-composition-properties-and-applications/) (https://texa-metals.com/blog/a360-aluminum-alloy-composition-properties-and-applications/)
21. [A360 Aluminum Alloy - Inox Cast - Inox Cast](https://www.inoxcast.com/a360-aluminum-alloy/) (https://www.inoxcast.com/a360-aluminum-alloy/)
22. [Aluminium LM24 - Teko](https://www.teko.it/uploads/manuali/materiali/LM_24.pdf) (https://www.teko.it/uploads/manuali/materiali/LM\_24.pdf)
23. [A380 Aluminum Alloy - oeform.com](https://oeform.com/die-cast-aluminium-alloy-a380-properties/) (https://oeform.com/die-cast-aluminium-alloy-a380-properties/)
24. [A356 Aluminum Alloy - wbcastings.com](https://www.wbcastings.com/non-ferrous-alloys/aluminum/a356/) (https://www.wbcastings.com/non-ferrous-alloys/aluminum/a356/)
25. [AlSi10Mg Alloy in Additive Manufacturing & Die Casting - LangHe Industry Co., Ltd. - langhe-industry.com](https://langhe-industry.com/alsi10mg-alloy/) (https://langhe-industry.com/alsi10mg-alloy/)
26. [Aluminum A357 Properties, Composition & Temper Guide - Metal Zenith - Metal Zenith](https://metalzenith.com/blogs/aluminium-properties/aluminum-a357-composition-properties-temper-guide-applications) (https://metalzenith.com/blogs/aluminium-properties/aluminum-a357-composition-properties-temper-guide-applications)
27. [ADC12 Aluminum - Redstone Manufacturing](https://redstonemanufacturing.com/ad12-aluminum/) (https://redstonemanufacturing.com/ad12-aluminum/)
28. [Die Casting vs. Permanent Mold - Batesville Products](https://batesvilleproducts.com/resource/die-casting-vs-permanent-mold/) (https://batesvilleproducts.com/resource/die-casting-vs-permanent-mold/)
29. [Reusable Aluminum Casting Molds: Permanent Mold, Steel Tooling, and When to Upgrade to Die Casting - casting-yz.com](https://casting-yz.com/reusable-aluminum-casting-molds-permanent-mold-steel-tooling-and-when-to-upgrade-to-die-casting/) (https://casting-yz.com/reusable-aluminum-casting-molds-permanent-mold-steel-tooling-and-when-to-upgrade-to-die-casting/)
30. [Aluminium Diecasting - LM6 - MRT Castings](https://www.mrt-castings.co.uk/aluminium-diecasting-lm6.html) (https://www.mrt-castings.co.uk/aluminium-diecasting-lm6.html)
31. [2021-NADCA-Aluminum.pdf - cwmdiecast.com](https://cwmdiecast.com/wp-content/uploads/2021/06/2021-NADCA-Aluminum.pdf) (https://cwmdiecast.com/wp-content/uploads/2021/06/2021-NADCA-Aluminum.pdf)
32. [Aluminum Alloy A356 Chemical Composition and Heat Treatment - oeform.com](https://oeform.com/aluminum-alloy-a356-chemical-composition-and-heat-treatment/) (https://oeform.com/aluminum-alloy-a356-chemical-composition-and-heat-treatment/)
33. [319 Aluminum Specifications and Alloy Characteristics - wbcastings.com](https://www.wbcastings.com/non-ferrous-alloys/aluminum/alloy-319/) (https://www.wbcastings.com/non-ferrous-alloys/aluminum/alloy-319/)
34. [Material - BS EN 1706 AC-46000 Copper Cu % 2.000 -4.000 Iron Fe % 1.300 max. - Gravity Cast India](https://gravitycastindia.com/images/resources/nonferrouspdf/BS%20EN%201706%20AC-46000.pdf) (https://gravitycastindia.com/images/resources/nonferrouspdf/BS%20EN%201706%20AC-46000.pdf)
35. [Norm: UNI EN 1676 e 1706 Numeric designation: EN AB and AC - 46000 - Raffmetal](https://www.raffmetal.com/wp-content/uploads/sites/4/EN-46000-AlSi9Cu3Fe-1.pdf) (https://www.raffmetal.com/wp-content/uploads/sites/4/EN-46000-AlSi9Cu3Fe-1.pdf)
36. [Microstructural Characteristics of AlSi9Cu3\(Fe\) Alloy with High Melting Point Elements - MDPI](https://www.mdpi.com/2075-4701/10/10/1278) (https://www.mdpi.com/2075-4701/10/10/1278)
37. [Data Sheet Aluminium AlSi10Mg Reference - xometry.pro](https://xometry.pro/wp-content/uploads/2023/08/Aluminium-1706-1.pdf) (https://xometry.pro/wp-content/uploads/2023/08/Aluminium-1706-1.pdf)
38. [NADCA\\_Alloy\\_Data\\_2009.pdf - tcdcinc.com](https://tcdcinc.com/assets/NADCA_Alloy_Data_2009.pdf) (https://tcdcinc.com/assets/NADCA\_Alloy\_Data\_2009.pdf)
39. [Aluminum LM24 - cass-usa.com](https://www.cass-usa.com/aluminum-lm24) (https://www.cass-usa.com/aluminum-lm24)
40. [AlSi10Mg Alloy Properties- Aluminum Die Casting - Sunrise Metal - Sunrise Metal](https://www.sunrise-metal.com/aluminum-alloy-alsi10mg/) (https://www.sunrise-metal.com/aluminum-alloy-alsi10mg/)

41. [A383 Aluminum - cass-usa.com](https://www.cass-usa.com/aluminum-a383-1) (<https://www.cass-usa.com/aluminum-a383-1>)
42. [The Difference Between Permanent, Sand Casting & Die Casting - LeClaire Manufacturing](https://www.leclairemfg.com/about-us/blog/difference-permanent-sand-casting-die-casting/) (<https://www.leclairemfg.com/about-us/blog/difference-permanent-sand-casting-die-casting/>)
43. [ASTM B85 / B85M - 14 Standard Specification for Aluminum-Alloy Die Castings - ASTM International](https://store.astm.org/b0085_b0085m-14.html) ([https://store.astm.org/b0085\\_b0085m-14.html](https://store.astm.org/b0085_b0085m-14.html))
44. [EOS Aluminium AlSi10Mg Material Data Sheet - EOS](https://www.eos.info/03_system-related-assets/material-related-contents/metal-materials-and-examples/metal-material-datasheet/aluminium/material_datasheet_eos_aluminium-alsi10mg_en_web.pdf) ([https://www.eos.info/03\\_system-related-assets/material-related-contents/metal-materials-and-examples/metal-material-datasheet/aluminium/material\\_datasheet\\_eos\\_aluminium-alsi10mg\\_en\\_web.pdf](https://www.eos.info/03_system-related-assets/material-related-contents/metal-materials-and-examples/metal-material-datasheet/aluminium/material_datasheet_eos_aluminium-alsi10mg_en_web.pdf))
45. [A360 Alloy Properties - Aluminum Die Casting - Sunrise Metal - Sunrise Metal](https://www.sunrise-metal.com/aluminum-alloy-a360/) (<https://www.sunrise-metal.com/aluminum-alloy-a360/>)
46. [Aluminum, Almen 308.0, Sand Cast :: MatWeb.com - MatWeb.com](https://www.matweb.com/search/datasheettext.aspx?matid=10253) (<https://www.matweb.com/search/datasheettext.aspx?matid=10253>)